CS 24 AP Computer Science A Review

Week 5: Writing Classes

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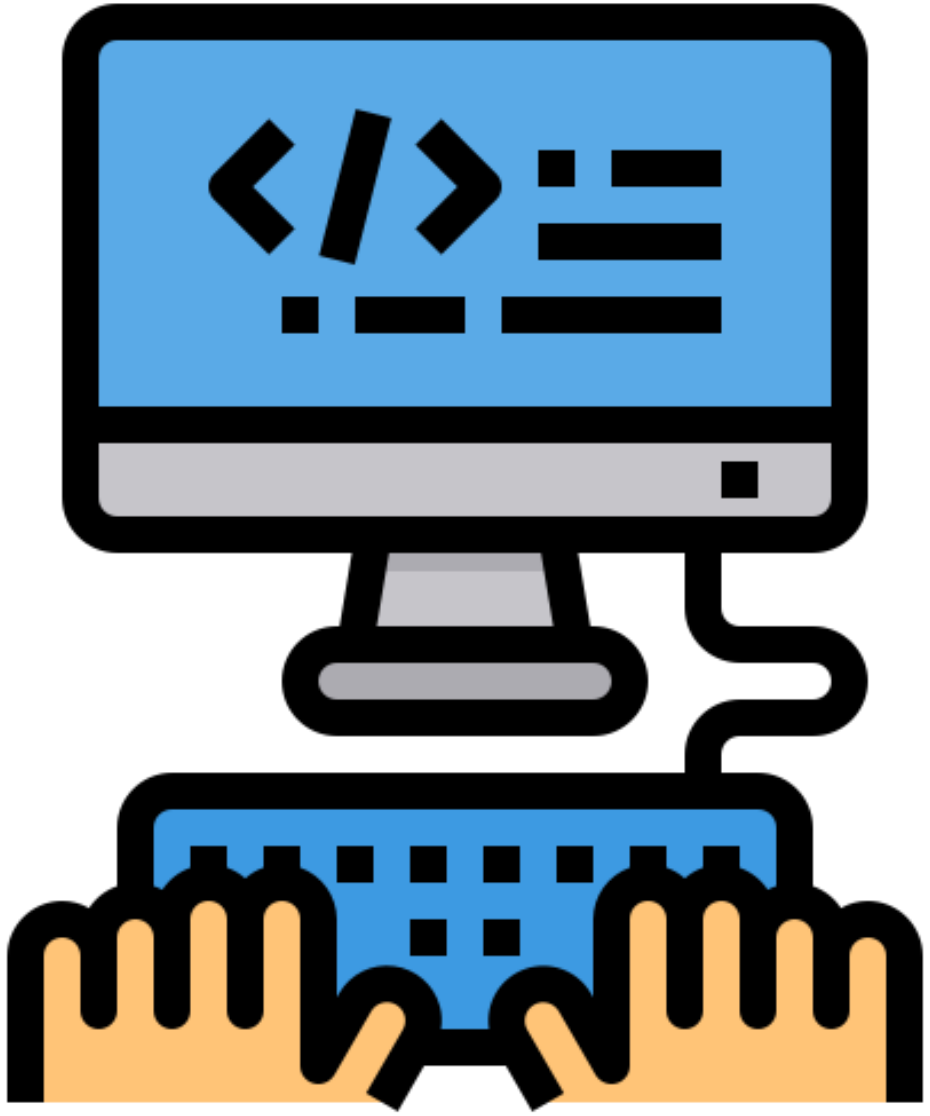
IEEE SENIOR MEMBER

Topics

- Design and Structure
- Methods
- Private/Public
- Object Creation (Composition), Reference and Object
- this Keyword
- Static versus Instance (Variables and Methods)



Class Design and Structure

SECTION 1

What is a class?

- A Class is a data type.
- A Class is the definition for the creation of objects.
- A Class is the template of objects
- A Class is a collection of data and methods
- A Class is the smallest program unit in Java language.

Basic Class Structure

```
public class GradePointAvg{
    // statements not shown
}
```

Class Types

- Main Program Class.
- Data Class
- Driver Class / Runner Class / Tester Class
- Library Class (Utility Class)
- Helper Class

Basic Class Structure

```
public class Circle{
     final static double PI = 3.14; // class variable
     double R;                      // instance variable
     Circle(double a){ R = a; }     // constructor
     public double getRadius(){ return R; }  // accessor
     public void setRadius(double a){   R = a; }  // mutator
     public double getArea(){ return PI * R *R; } // derived data
     static double getArea(Circle c){  return PI * c.R * c.R; }
                                                  // static method
     // overridden methods (Object Class)
     public String toString(){  return "("+R+")"; }
     public Boolean equals(Object c){ return this.R == c.R; }
}
```

Multiple Choice (1)

Which of the following class declarations would cause a compile-time error? Assume the rest fo the code compiles as intended.

Multiple Choice (1)

Ⓐ public class Calculus

Ⓑ public class apCalculus

Ⓒ public class APCalculus

Ⓓ public class 4APCalculus

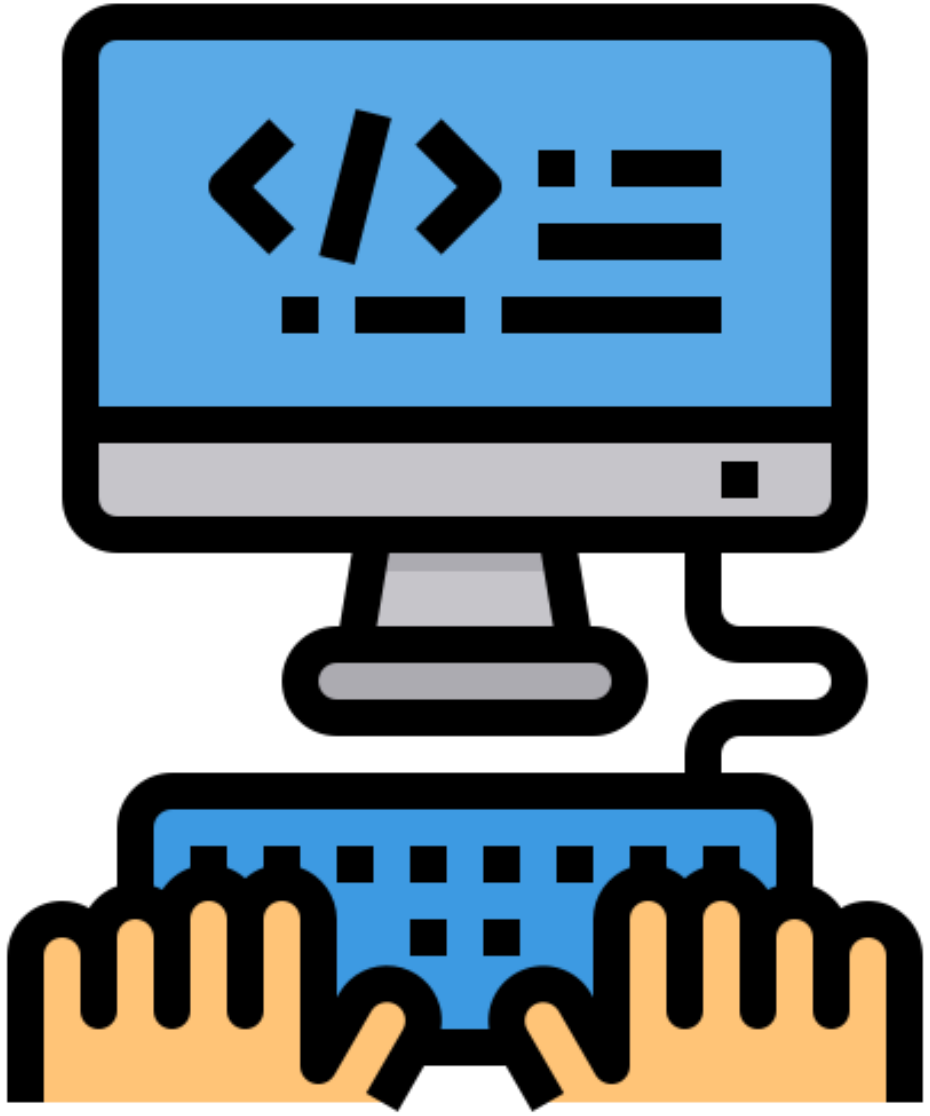
Multiple Choice (1)

Ⓐ public class Calculus

Ⓑ public class apCalculus

Ⓒ public class APCalculus

Ⓓ public class 4APCalculus

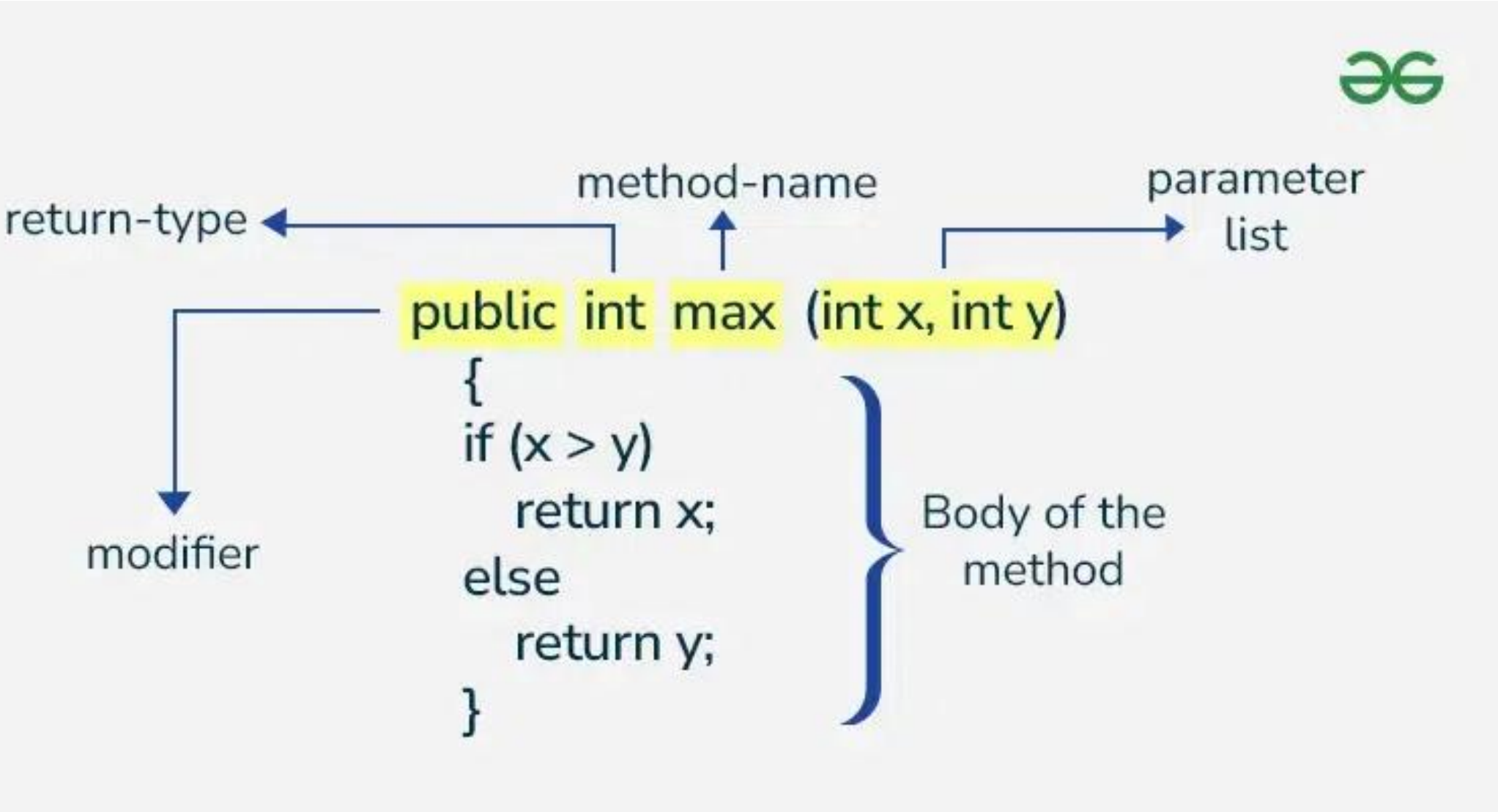


Methods

SECTION 2

Methods

- A Method is a group of cod that performs a specific task. (function)
- Class Method (static), Instance Method (non-static)
- Procedure (Function without return value)
- Function (Function with return value)
- Method (Function in Object-Oriented Programming)


return-type
method-name
parameter list
public int max (int x, int y)
{
if (x > y)
return x;
else
return y;
}
modifier
Body of the method

Overloading

- Methods of the same class with the same name but different parameter list.

```
float add(int a, float b);
float add(int a, float b, int c);
float add(float a, int b);
float add(float a, float b);
```

Valid Overloading signatures

```
int abc(int num);
float abc(int num);
```

Invalid Overloading signatures

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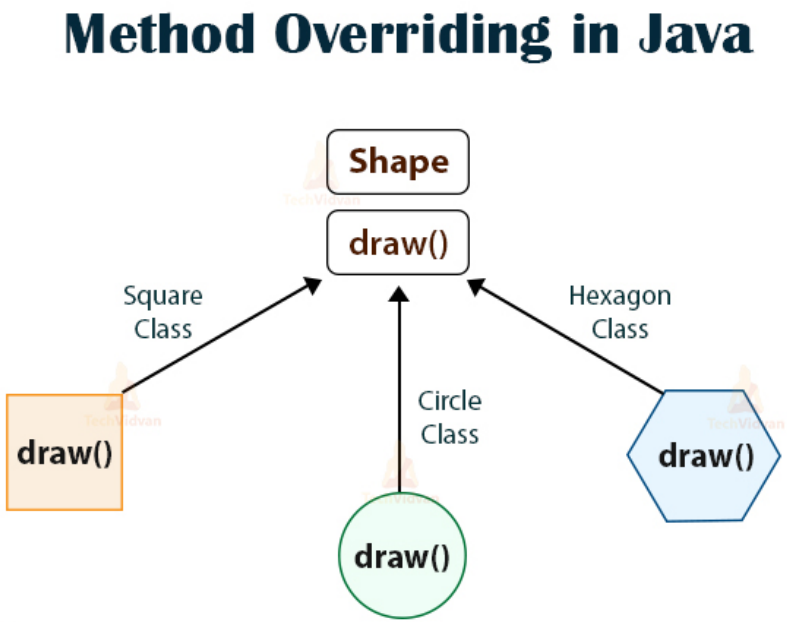
Overriding

- Methods of the same method and parameter list for parent-child classes.



Multiple Choice (2)

Which of the following statements would best describe an efficient design to represent a pair of sunglasses?

Multiple Choice (2)

Ⓐ Three classes: UnfoldGlasses, CleanGlasses, and WearGlasses

Ⓑ An UnfoldGlasses class with methods CleanGlasses and WearGlasses

Ⓒ A PairOfSunglasses class with Boolean variables unfolded, cleaned, and worn

Ⓓ A PairOfSunglasses class with methods that unfold, clean, and wear the objects in the class.

Multiple Choice (2)

Ⓐ Three classes: UnfoldGlasses, CleanGlasses, and WearGlasses

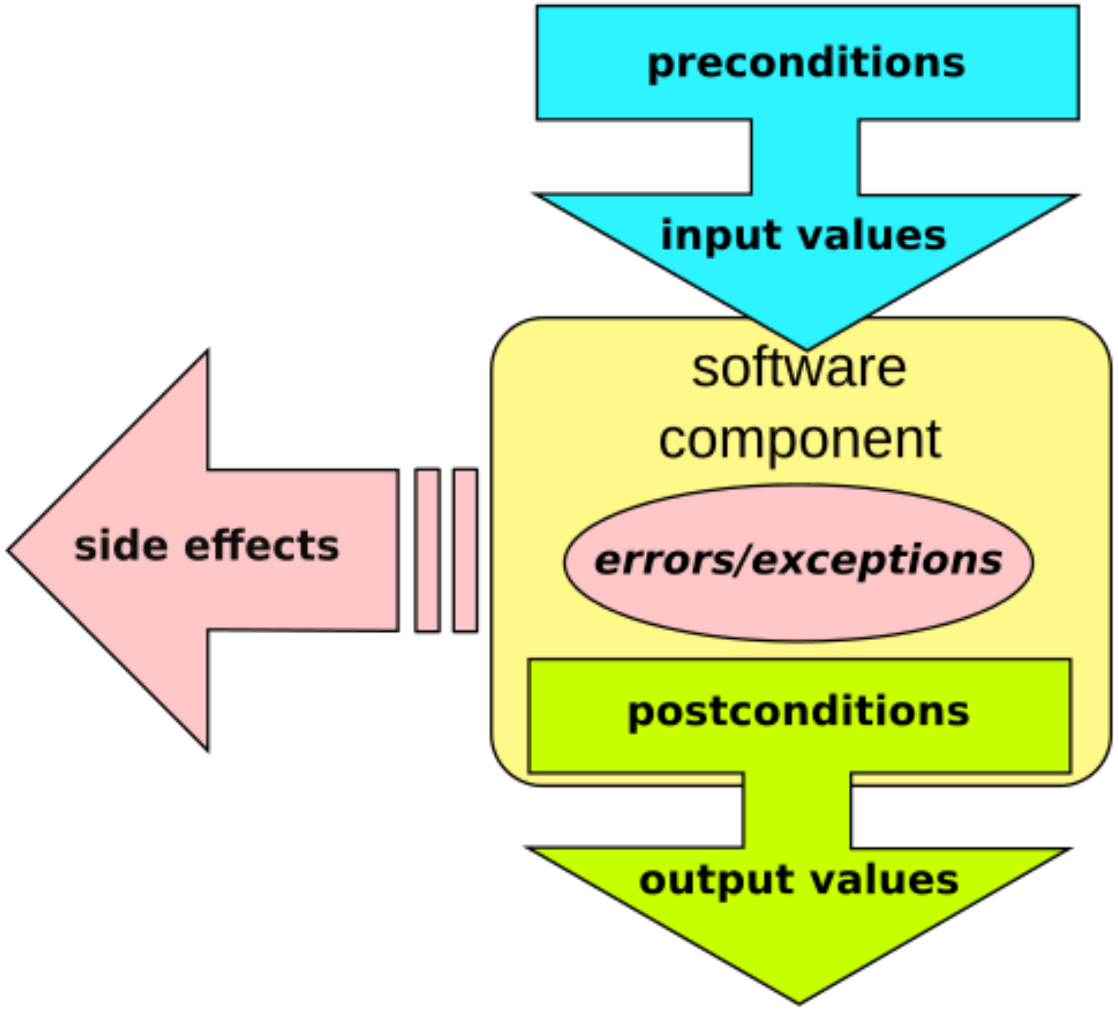
Ⓑ An UnfoldGlasses class with methods CleanGlasses and WearGlasses

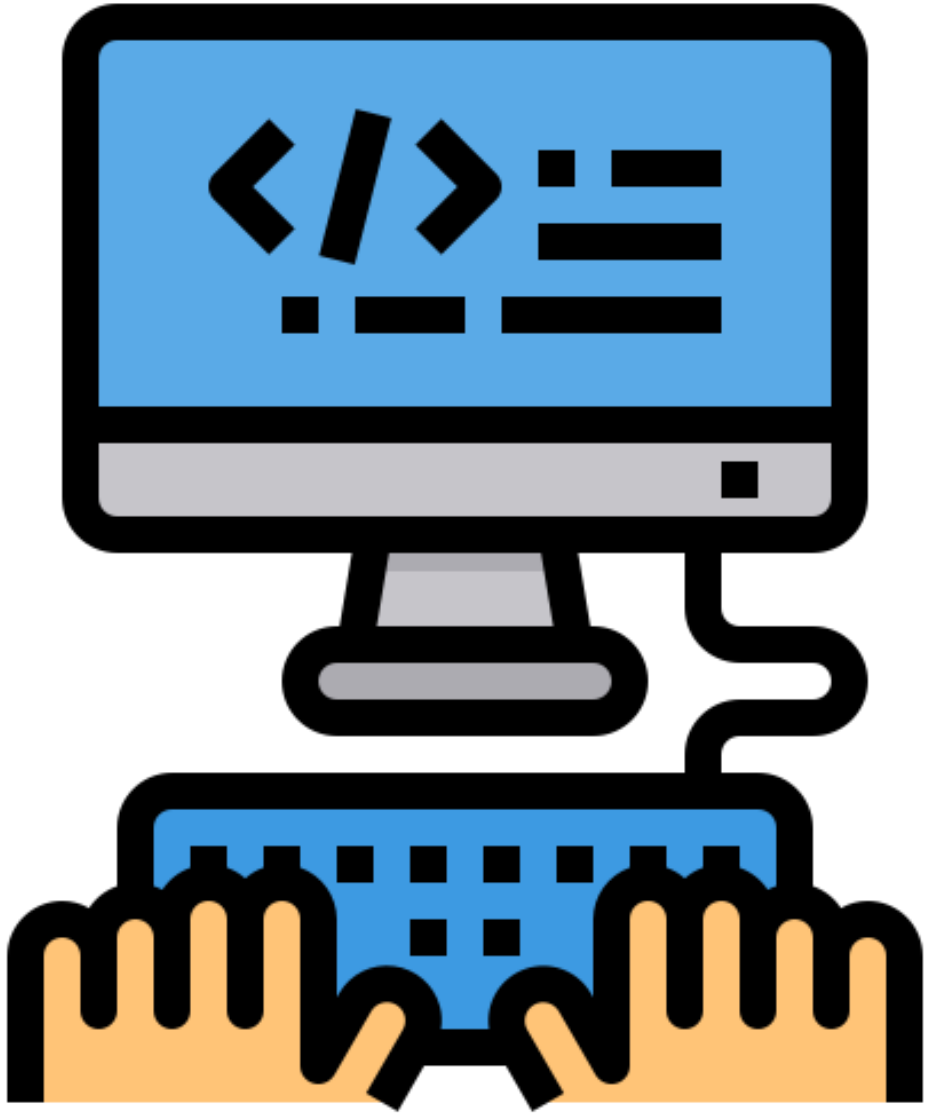
Ⓒ A PairOfSunglasses class with Boolean variables unfolded, cleaned, and worn

Ⓓ A PairOfSunglasses class with methods that unfold, clean, and wear the objects in the class.

Note:

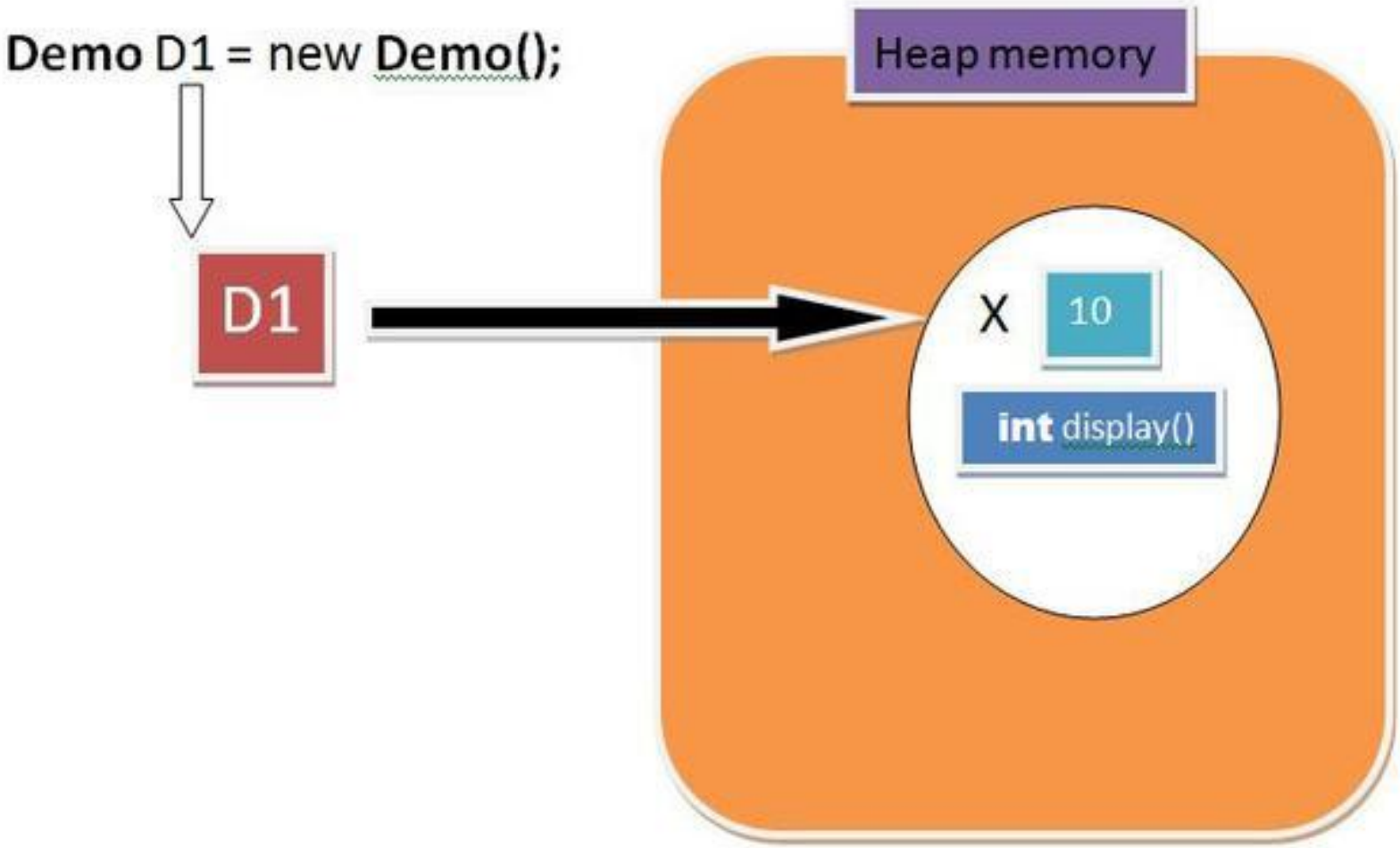
- Nouns -> variables.
- Verb -> Methods

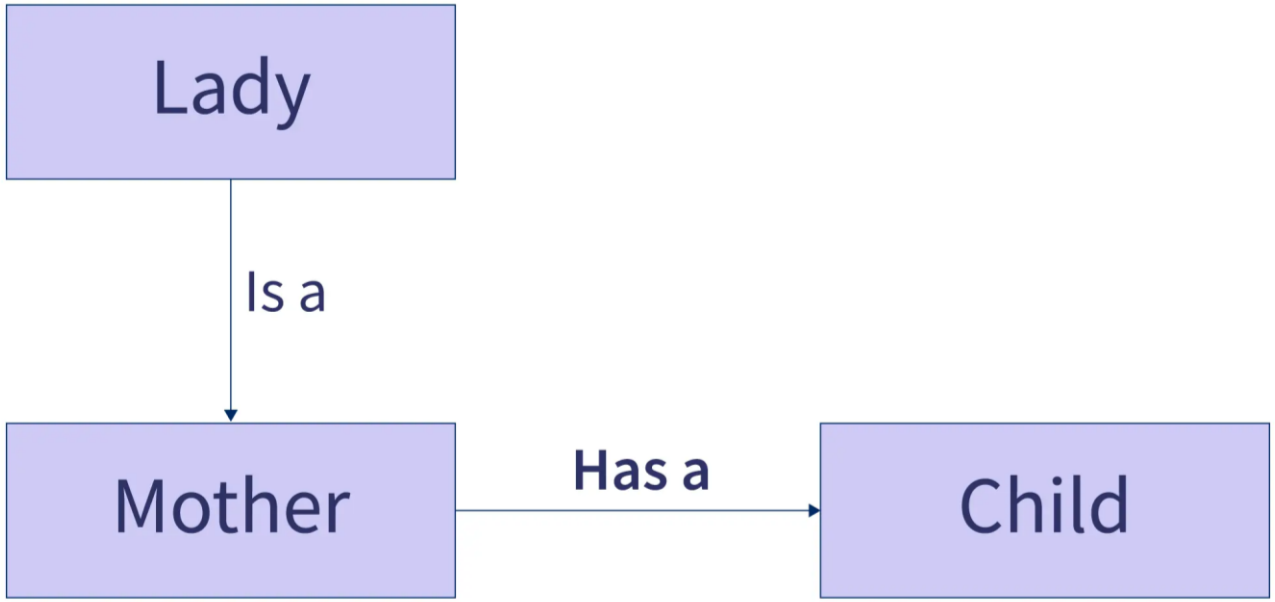

preconditions
input values
software
component
side effects
errors/exceptions
postconditions
output values



Object and Reference

SECTION 3


Demo D1 = new Demo();
D1
Heap memory
X
10
int display()



SCALER Topics

```
class A {
  .......
}
class B
{

 void disp()
 {
  A obj=new A();
  ........
  ........
 }
}
```

Uses-A Relationship

```
class A {
  .......
}
class B
{
  A obj=new A();
  ........
  ........
}
```

Has-A Relationship

```
class A {
  ........
}
class B extends A
{
  ........
  ........
}
```

Is-A Relationship

Fig: Different forms of relationship between classes in Java

Multiple Choice (3/4)

```
public class Chair{
    private int numberOfLegs = 4;
    private boolean padded;
    public Chair(boolean soft){
        if (soft) padded = true;
        else padded = false;
    }
}
// In another class.
public void sitOnChair(){ <implementation not
shown>}
```

Multiple Choice (3)

The method sitOnChair belongs to another class and is supposed to allow the user to “sit” on a Chair if the chair is padded. Which of the following code segments could be used to replace <implementation not shown> so that sit OnChair will work as intended?

Ⅰ.
```
Chair c = new Chair(true);
c.sit();
```

Ⅱ.
```
Chair c = new Chair(true);
```

Ⅲ.
```
Chair c = new Chair(true);
if (c.padded) System.out.println(“You are sitting”);
```

Multiple Choice (3)

Ⓐ Ⅱ only

Ⓑ Ⅲ only

Ⓒ Ⅰ, Ⅱ and Ⅲ

Ⓓ None

Multiple Choice (3)

Ⓐ Ⅱ only

Ⓑ Ⅲ only

Ⓒ Ⅰ, Ⅱ and Ⅲ

Ⓓ None

Multiple Choice (4)

Which of the following modifications, if any, would help to make the Chair class Most useful to the `sitOnChair` method, based on the task attempted in Question 3?

Multiple Choice (4)

Ⓐ Adding an accessor method that return the value of numberOfLegs

Ⓑ Adding an accessor method that returns the value of padded

Ⓒ Adding a mutator that changes the value of numberOfLegs

Ⓓ Adding a mutator method that changes the value of padded.

Multiple Choice (4)

Ⓐ Adding an accessor method that return the value of `numberOfLegs`

Ⓑ Adding an accessor method that returns the value of `padded`

Ⓒ Adding a mutator that changes the value of `numberOfLegs`

Ⓓ Adding a mutator method that changes the value of `padded`.

Multiple Choice (5)

Consider the following method:

```
public int halfRoot(int n){
    return Method.sqrt(n)/2;
}
```

Multiple Choice (5)

Ⓐ `halfRoot(-2)`

Ⓑ `halfRoot((int) 2.0)`

Ⓒ `halfRoot(3.0)`

Ⓓ None will cause run-time error.

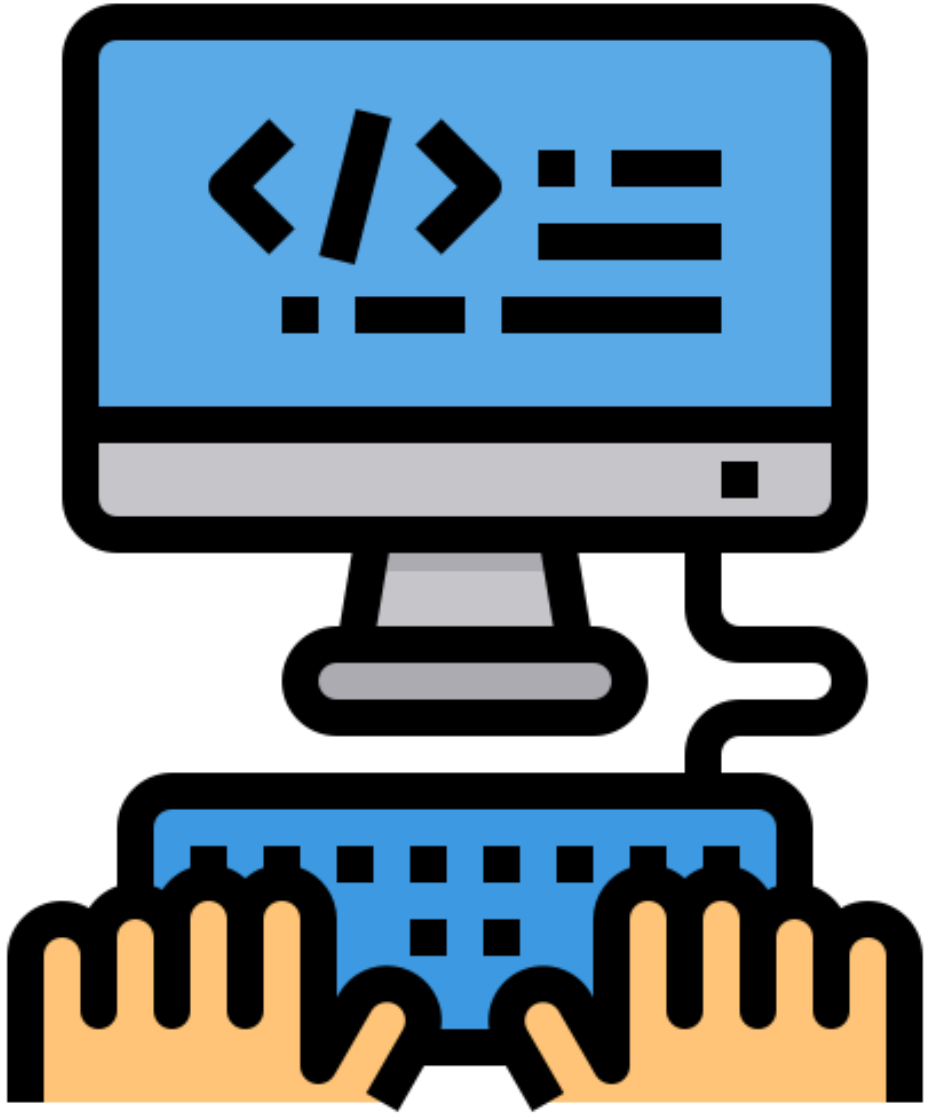
Multiple Choice (5)

Ⓐ `halfRoot(-2)`

Ⓑ `halfRoot((int) 2.0)`

Ⓒ `halfRoot(3.0)`

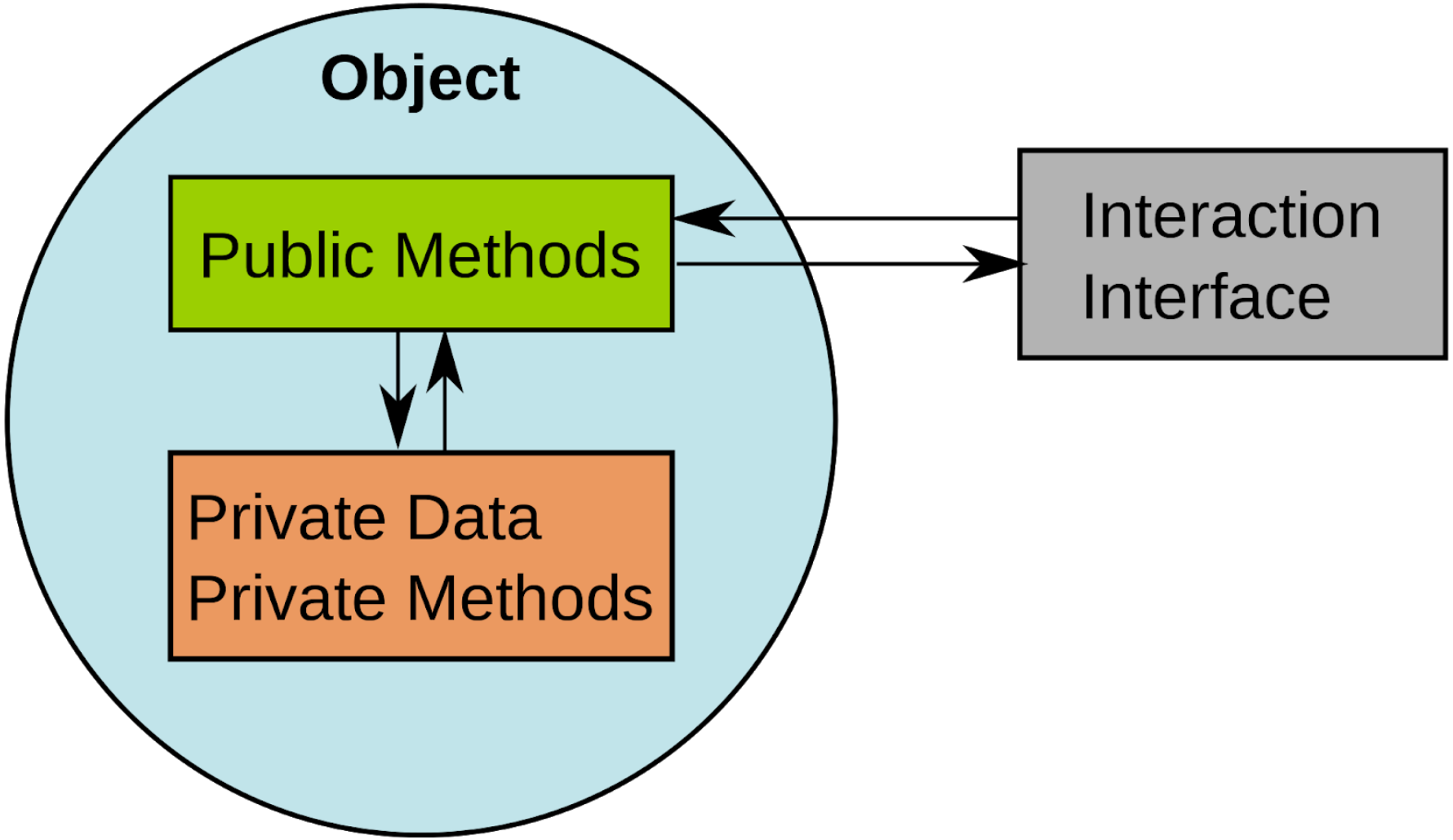
Ⓓ None will cause run-time error.



Encapsulation – private/public

SECTION 4

Modifier	Class	Package	Subclass	World
`public`	✔	✔	✔	✔
`protected`	✔	✔	✔	✘
`no modifier*`	✔	✔	✘	✘
`private`	✔	✘	✘	✘


Object
Public Methods
Private Data
Private Methods
Interaction
Interface

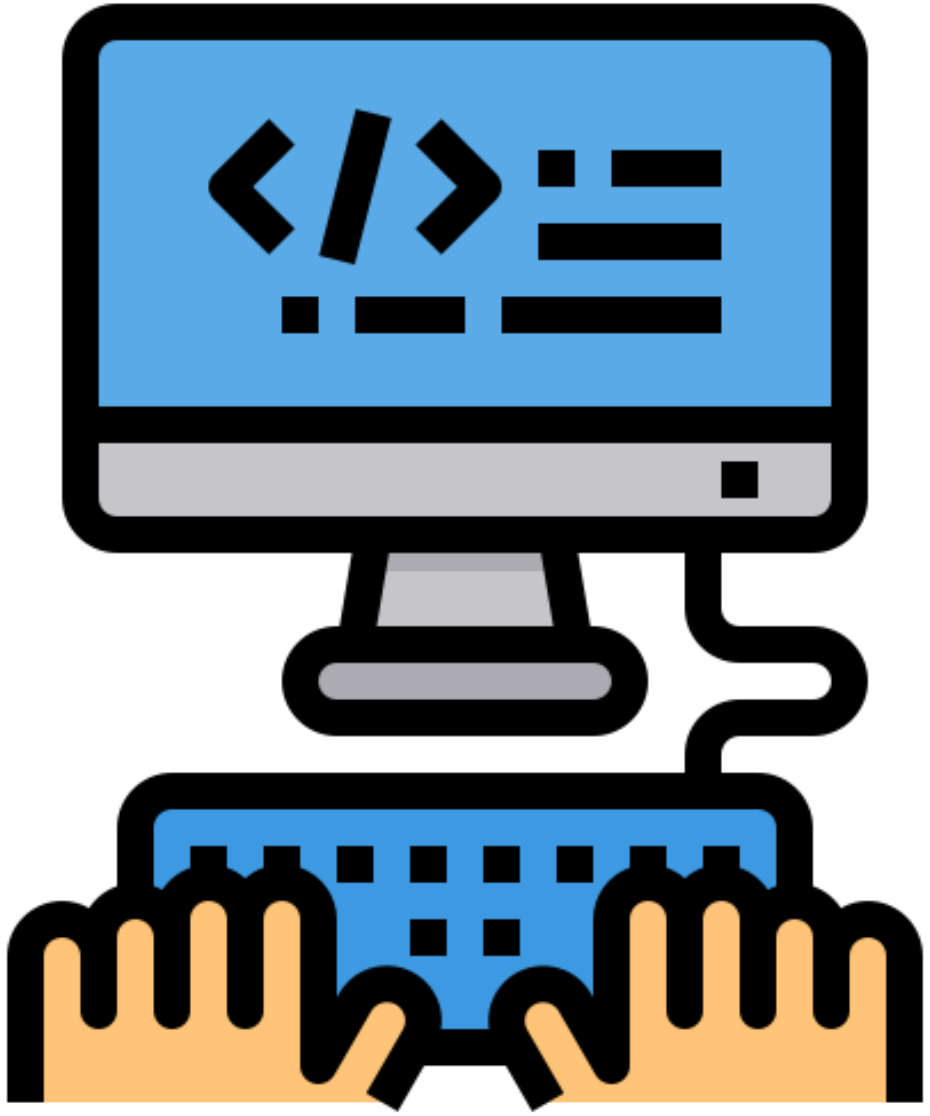
```
public class Student{

 private int age;

 public void setAge(int studentAge){
        this.age = studentAge;
      }

      public int getAge() {
        return this.age;
      }
}
```

Methods and variables are memebers of Class.

Members of Capsule = White Ball

Class = Single Unit

1 capsule = Single Unit



This/Super Keyword

SECTION 5

this Keyword in Java

Invoking an overloaded constructor using **this** keyword

```java
public class Circle {

    private Point center;
    private double radius;

    public Circle() {
        this(new Point(1, 1), 1);
    }

    public Circle(Point center, double radius) {
        this.center = center;
        this.radius = radius;
    }

    ...
}
```

118 Java

Super Keyword in Java

1. To Refer Instance Variable of an Immediate Parent Class
2. To Refer Method of An Immediate Parent Class
3. To Refer Constructor of Immediate Parent Class

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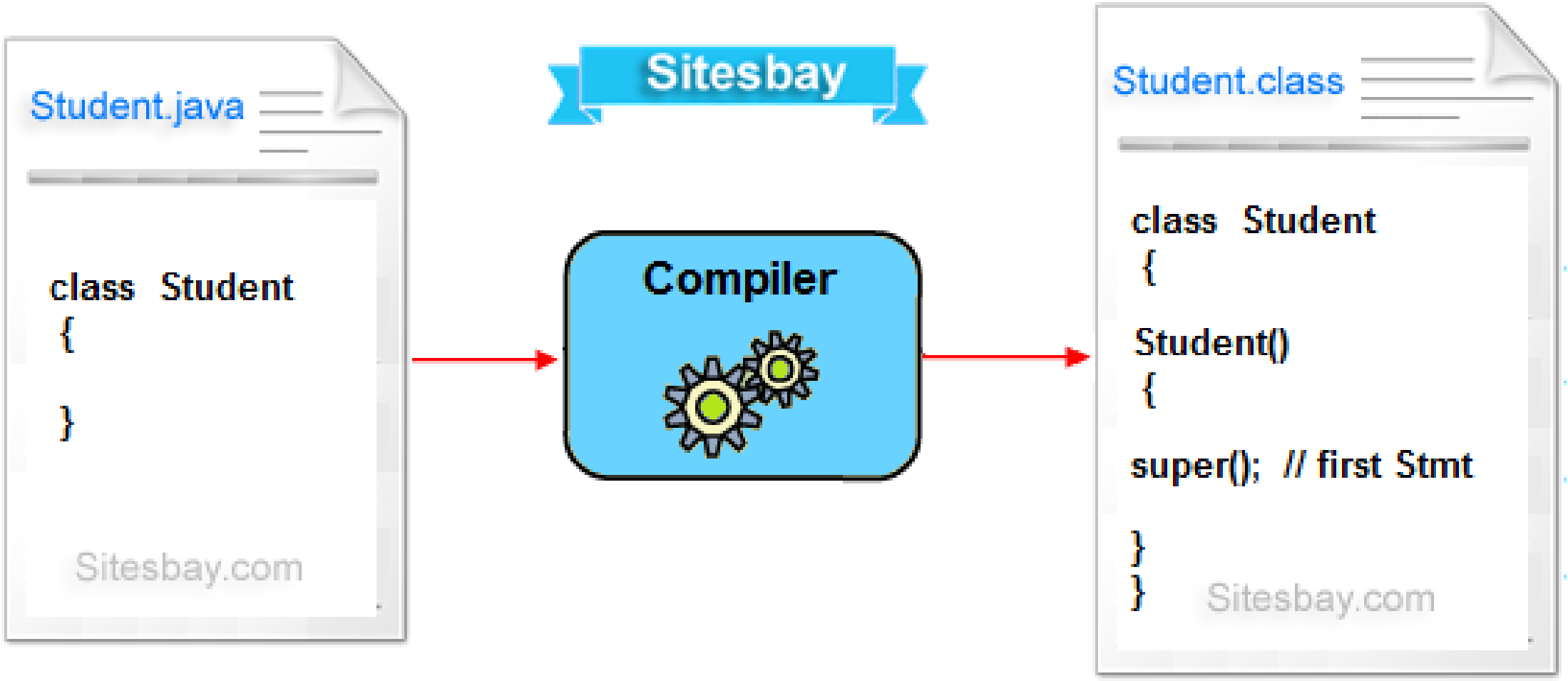

Student.java
class Student
{
}
Sitesbay.com
Sitesbay
Compiler
Student.class
class Student
{
Student()
{
super(); // first Stmt
}
}
Sitesbay.com

```
class Person {
    Person() {
    }
}
```

Parent Class

super()

```
class Student
  extends Person {
    Student() {
      super();
    }
}
```

Child Class

Multiple Choice (6)

The `changeVal` method in `Number` should reassign `value` to be the value taken as the parameter and return the original value. Which of the following code segments should be used to replace /* missing code */ so that `changeVal` will work as intended?

Multiple Choice (6)

Ⓐ
```
value = new Val;
return value;
```

Ⓑ
```
int oldVal = value;
value = newVal;
```

Ⓒ
```
int oldVal = value;
value = newVal;
return value;
```

Ⓓ
```
int oldVal = value;
value = newVal;
return oldVal;
```

Multiple Choice (6)

Ⓐ
```
value = new Val;
return value;
```

Ⓑ
```
int oldVal = value;
value = newVal;
```

Ⓒ
```
int oldVal = value;
value = newVal;
return value;
```

Ⓓ
```
int oldVal = value;
value = newVal;
return oldVal;
```

Multiple Choice (7)

Which will be printed as a result of executing the `change` method? Assgume `changeVal` in the `Number` class works as intended?

Multiple Choice (7)

Ⓐ 5 5

Ⓑ 5 10

Ⓒ 10 10

Ⓓ None of these.

Multiple Choice (7)

Ⓐ 5 5

Ⓑ 5 10

Ⓒ 10 10

Ⓓ None of these.

Multiple Choice (8)

The speed limit of a stretch of highway is 55 miles per hour (mph). The highway patrol issues speeding tickets to anyone caught going faster than 55 miles per hour. The fine for speeding is based on the following scale:

Speed	**Fine**
greater than 55 mph but less than 65 mph	$100
greater than or equal to 65 mph but less than 75 mph	$150
greater than or equal to 75 mph	$300

Multiple Choice (8)

If the value of `int` variable `speed` is the speed of a driver who was pulled over for going faster than 55 mph, which of the following code segments will assign the correct value to the int variable `fine`?

```
I. if (speed >=75)fine = 300;
   if (speed >=65 && speed < 75)fine = 150;
   if (speed > 55 && speed <65) fine = 100;

II. if (speed >=75)fine = 300;
    if (65 <= speed < 75)fine = 150;
    if (55 < speed < 65) fine = 100;
III. if (speed >= 75) fine = 300;
    if (speed >= 65) fine = 150;
    if (speed > 55) fine = 100;
```

Multiple Choice (8)

Ⓐ I only

Ⓑ I and Ⅲ only

Ⓒ Ⅱ and Ⅲ only

Ⓓ I , Ⅱ and and Ⅲ

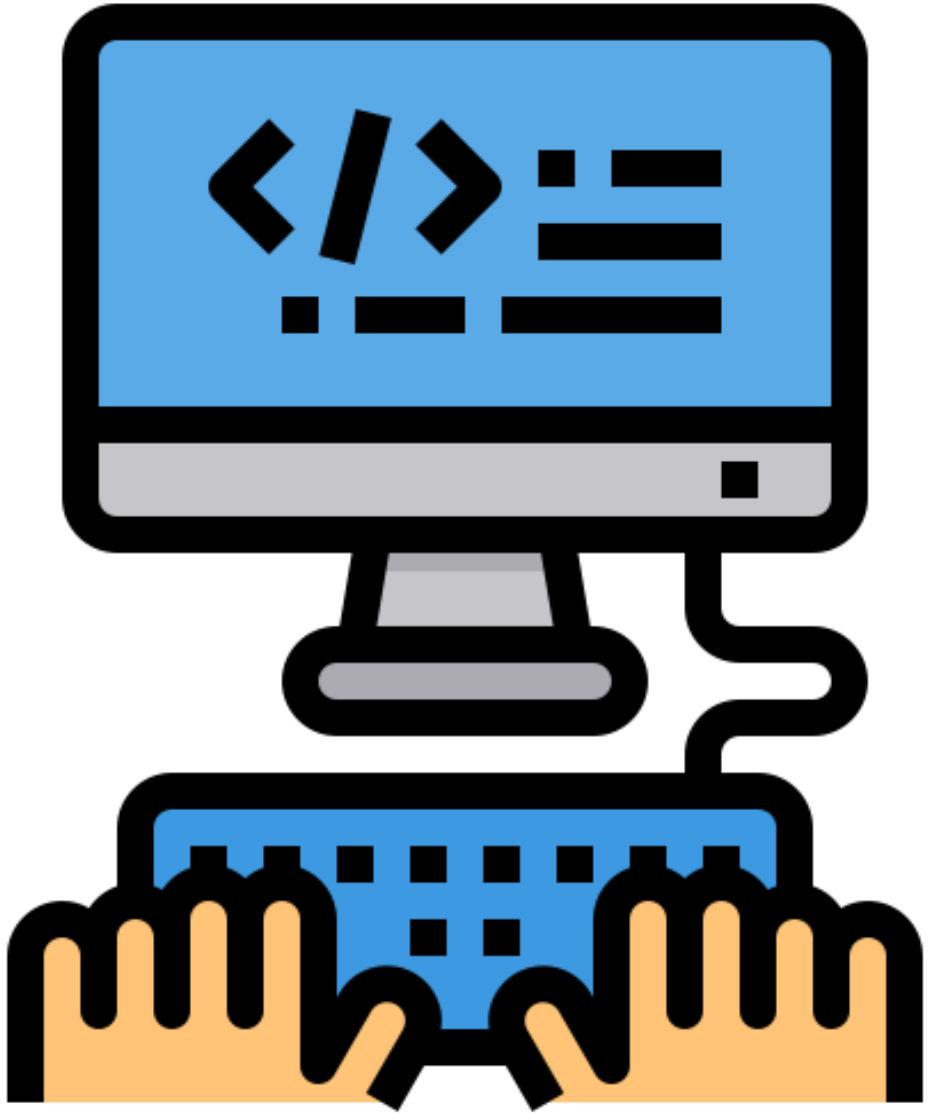
Multiple Choice (8)

Ⓐ I only

Ⓑ I and Ⅲ only

Ⓒ Ⅱ and Ⅲ only

Ⓓ I , Ⅱ and and Ⅲ

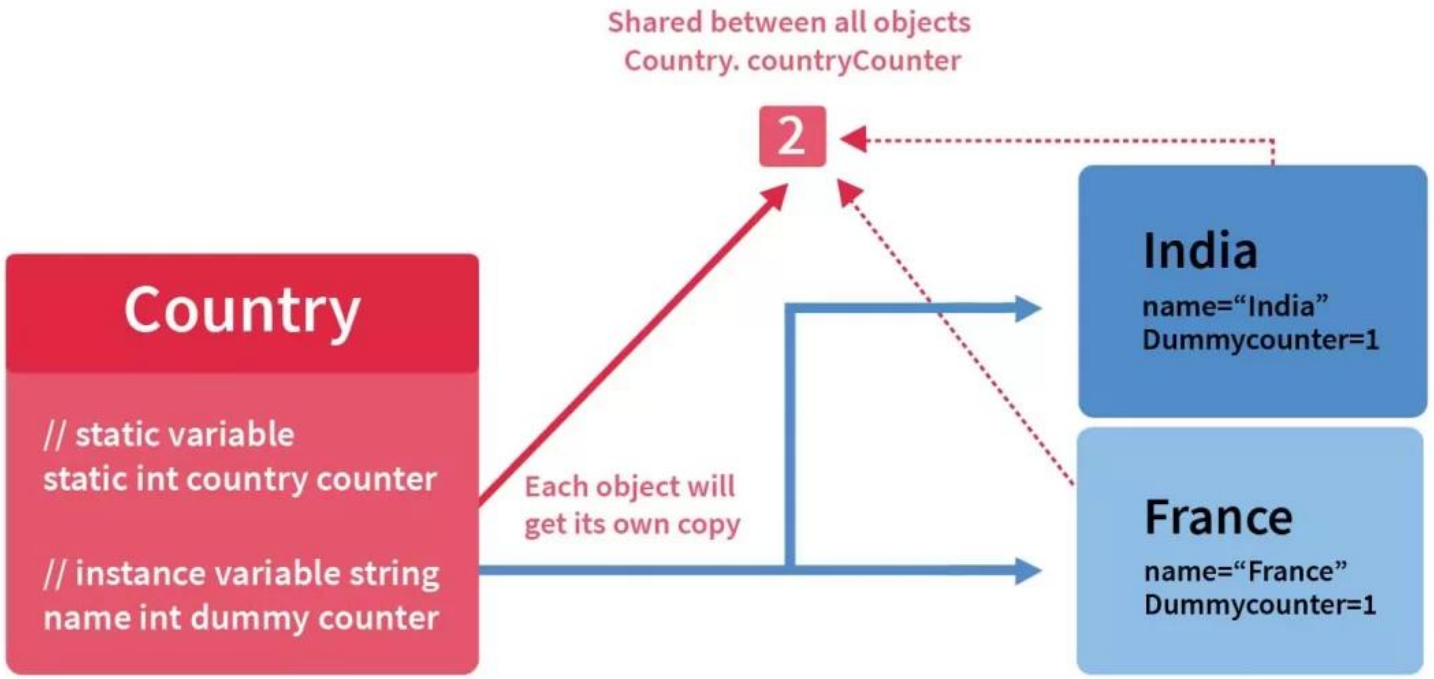


Static versus Instance variables and Methods

SECTION 6

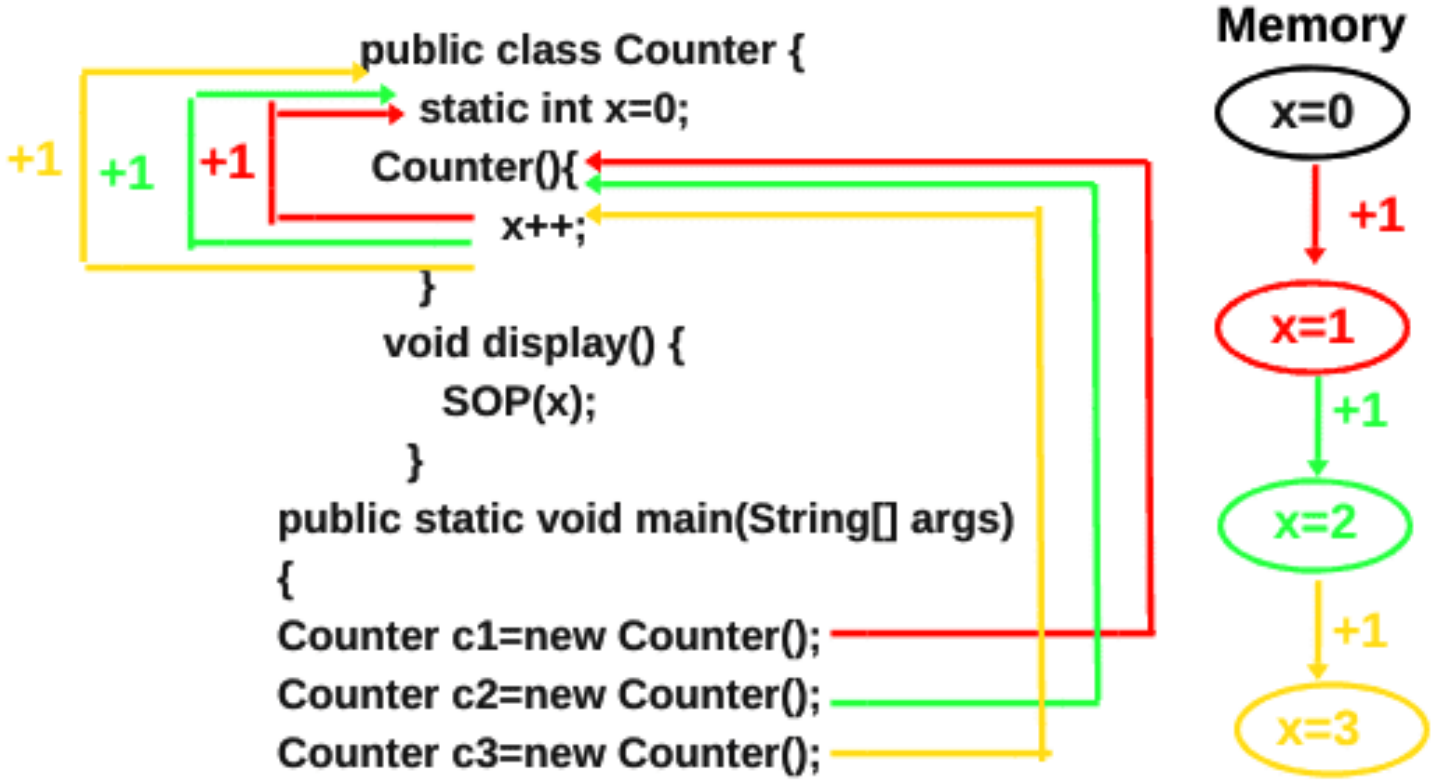
Type of Variable

- Local Variables
- Instance Variables
- Static Variables



SCALER Topics

```
public class Counter {
    static int x=0;
    Counter(){
        x++;
    }
    void display() {
        SOP(x);
    }
public static void main(String[] args)
{
Counter c1=new Counter();
Counter c2=new Counter();
Counter c3=new Counter();
    c1.display();
    c2.display();
    c3.display(); }
}
```



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Difference Between Non-static and Static Method

Non-static Method	Static Method
• It is specific to an object so that these are also known as instance method. • These methods always access with object reference **Syntax:**	• These are common to every object so that it is also known as member method or class method. • These property always access with class reference **Syntax:**
Objref.methodname()	**className.methodname();**
• If any method wants to be execute multiple time that can be declare as non static.	• If any method wants to be execute only once in the program that can be declare as static .

Multiple Choice (9)

A cla ss called `ComputerMouse` has a static variable `connector` and a static method `getConnector`. Which of the statements is true based on this information?

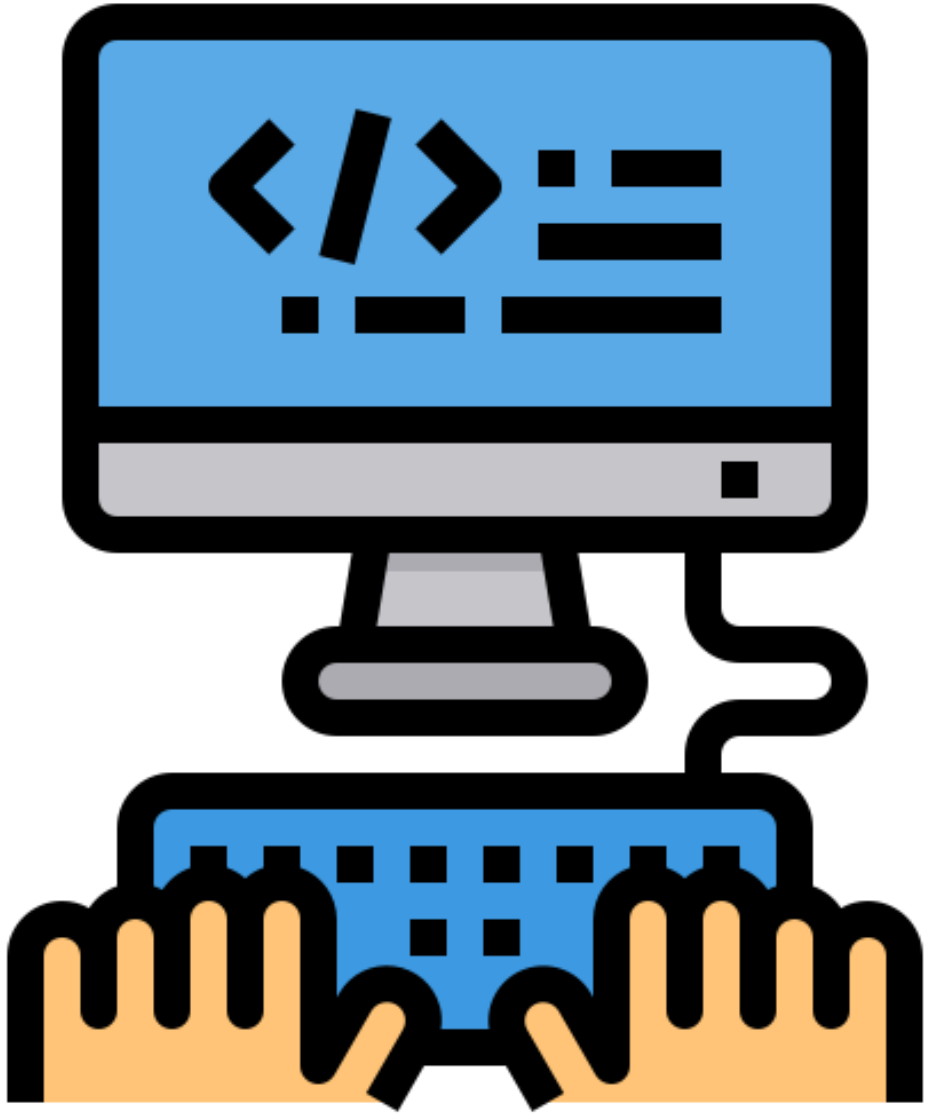
Multiple Choice (9)

Ⓐ In order to invoke `getConnector`, a new `ComputerMouse` object does not need to be instantiated; `getConnector` must be called directly through the object class.

Ⓑ In order to invoke `getConnector`, a new `ComputerMouse` object must be instantiated and the `getConnector` must be calle thorugth that object.

Ⓒ Since `connector` is declared static, `getConnector` is shared among all objects in the program.

Ⓓ Since `connector` is declared static, all of the methods in `ComputerMouse` must also be declared static.



Practice MCQ Questions

SECTION 5

Multiple Choice (1)

A development team is building an online bookstore that customers can sue to order books. Information about inventory and customer orders is kept in a data base. Code must be developed that will store and retrieve data from the data base. The development team decides to put the database code in separate classes from the rest of the program. Which of the following would be an advantage of this plan?

Multiple Choice (1)

Ⅰ. The database access code could be reused in other applications that also need to access the database.

Ⅱ. The database access code can be tested independently. It will be possible to test the database access code before the interface is developed.

Ⅲ. A team of programmers can be assigned to work just on the code that is used to access the database. The programmers can work independently from other programmers, such as those who develop the user interface

Multiple Choice (1)

Ⓐ Ⅰ only

Ⓑ Ⅲ only

Ⓒ Ⅰ and Ⅱ

Ⓓ Ⅰ , Ⅱ and Ⅲ

Multiple Choice (1)

Ⓐ Ⅰ only

Ⓑ Ⅲ only

Ⓒ Ⅰ and Ⅱ

Ⓓ Ⅰ , Ⅱ and Ⅲ

Multiple Choice (2)

In Java. instance fields (also known as instance variables) and methods can be designated public or private. Which of the following best characterizes the designation that should be used?

Multiple Choice (2)

Ⓐ Instance field and methods should always be public. This makes it easier for client programs to access data fields and use the methods of the class.

Ⓑ Instance fields should be either public or private, depending on whether or not it is beneficial for client programs to access them directly. All methods should be public. A private method is useless because a client program can't access it.

Ⓒ Keep all methods and instance fields private. This enforce encapsulation.

Ⓓ Instance fields should always be private so that clients can't directly access them. Methods can be either public or private.

Multiple Choice (2) ?

Ⓐ Instance field and methods should always be public. This makes it easier for client programs to access data fields and use the methods of the class.

Ⓑ Instance fields should be either public or private, depending on whether or not it is beneficial for client programs to access them directly. All methods should be public. A private method is useless because a client program can't access it.

Ⓒ Keep all methods and instance fields private. This enforce encapsulation.

Ⓓ Instance fields should always be private so that clients can't directly access them. Methods can be either public or private.

Multiple Choice (3)

Which of the following are signs of a well-designed program?

I. Clients know how data is stored in the class.

II. Classes and methods can be tested independently.

III. The implementation of a method can be changed without changing the programs that use the method.

Multiple Choice (3)

Ⓐ Ⅰ only

Ⓑ Ⅱ only

Ⓒ Ⅱ and Ⅲ

Ⓓ Ⅰ and Ⅱ

Multiple Choice (3)

Ⓐ Ⅰ only

Ⓑ Ⅱ only

Ⓒ Ⅱ and Ⅲ

Ⓓ Ⅰ and Ⅱ

Multiple Choice (4)

Consider the following classes:

```
public class Sample{
    public void writeMe(Object obj){
      System.out.println("Object");
    }
    public void writeMe(String s){
      System.out.println("string");
    }
}
```

Multiple Choice (4)

What will be the result of executing the folliwing?

```
Sample s = new Sample();
String tmp = new String("hi");
s.writeMe(tmp);
```

Multiple Choice (4)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (4)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (5)

Consider the following classes:

```
public class Sample{
    public void writeMe(Object obj){
      System.out.println("Object");
    }
    public void writeMe(String s){
      System.out.println("string");
    }
}
```

Multiple Choice (5)

What will be the result of executing the folliwing?

```
Sample s = new Sample();
Object tmp = new Object();
s.writeMe(tmp);
```

Multiple Choice (5)

Ⓐ Compile-time error.

Ⓑ “string”.

Ⓒ “object”.

Ⓓ Run-time error.

Multiple Choice (5)

Ⓐ Compile-time error.

Ⓑ “string”.

Ⓒ “object”.

Ⓓ Run-time error.

Multiple Choice (6)

Consider the following classes:

```
public class Sample{
    public void writeMe(Object obj){
      System.out.println("Object");
    }
    public void writeMe(String s){
      System.out.println("string");
    }
}
```

Multiple Choice (6)

What will be the result of executing the folliwing?

```
Sample s = new Sample();
Object tmp = new String("hi");
s.writeMe(tmp);
```

Multiple Choice (6)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (6)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (7)

Consider the following classes:

```
public class Sample{
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      System.out.println("Object");
    }
    public void writeMe(String s){
      System.out.println("string");
    }
}
```

Multiple Choice (7)

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```
Sample s = new Sample();
String tmp = new Object();
s.writeMe(tmp);
```

Multiple Choice (7)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (7)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

Multiple Choice (8)

Consider the following class code:

```
public class sample{
    int val = 0;
}
```

Is `val` an attribute or a method?

Multiple Choice (8)

Ⓐ `val` is an attribute.

Ⓑ `val` is a method.

Ⓒ `val` is both an attribute and a method.

Ⓓ Neither : `val` is a primitive.

Multiple Choice (8)

Ⓐ `val` is an attribute.

Ⓑ `val` is a method.

Ⓒ `val` is both an attribute and a method.

Ⓓ Neither : `val` is a primitive.

Multiple Choice (9)

Consider the following classes:

```
public class Sample{
    public void writeMe(Object obj){
      System.out.println("Object");
    }
    public void writeMe(String s){
      System.out.println("string");
    }
}
```

Multiple Choice (9)

What will be the result of executing the folliwing?

```
Sample s = new Sample();
Object tmp = new Object();
s.writeMe(tmp);
```

Multiple Choice (9)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.

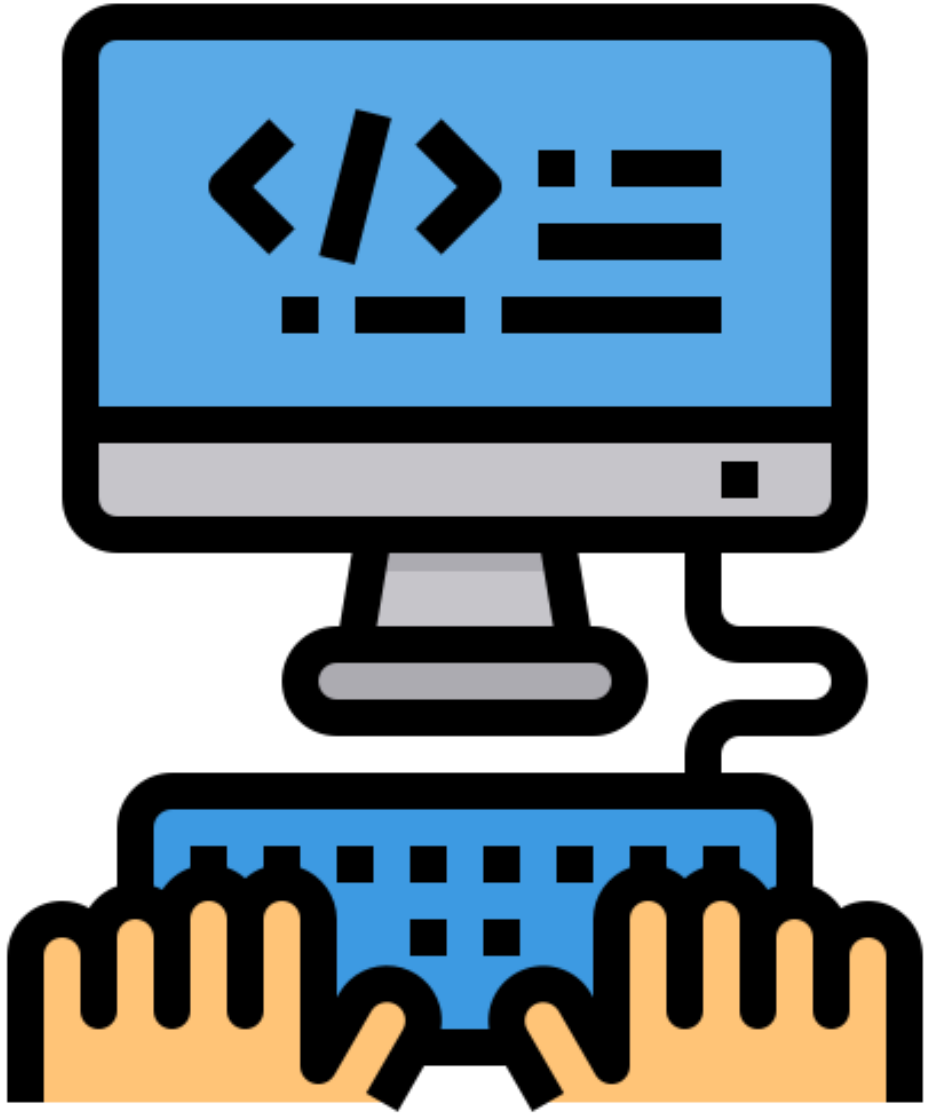
Multiple Choice (9)

Ⓐ Compile-time error.

Ⓑ “object”.

Ⓒ “string”.

Ⓓ Run-time error.



Class Creation

SECTION 6

Solving the Class Design Problems

Must understand the problem and keep notice of the following issues:

1. Class Name
2. Extension?
3. Instance fields
4. static fields
5. Constants
6. Instance methods
7. Class Methods.
8. Derived data fields
9. Overloading and Overriding.